

Quiz: Recursion (Makeup)

Total Score: $/2^2$

Printed Name:

Quiz rules:

1. You MAY use any printed or handwritten notes.
2. You MAY NOT use a computer or any other electronic device.

Problem 1. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 0:
5         return 0
6     ret = foo(xs[1:])
7     return xs[0] + ret * 3
8 try:
9     xs = [1, 2, 3, 4, 5]
10    print('foo(xs)=',foo(xs))
11 except RuntimeError:
12    print('StackOverflow')
13 EOF
14 $ python3 foo.py
```

Fraction of LLMs with correct answer: $9 / 16 = 0.56$

Problem 2. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 0:
5         return 1
6     return xs[0] + foo(xs[1:]) + foo(xs[3:])
7 try:
8     print('foo([1, 2, 3, 4])=',foo([1, 2, 3, 4]))
9 except RuntimeError:
10    print('StackOverflow')
11 EOF
12 $ python3 foo.py
```

Fraction of LLMs with correct answer: $10 / 16 = 0.62$

Problem 3. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def is_even(n):
4     if n == 0:
5         return True
6     return is_odd(n - 1)
7
8 def is_odd(n):
9     if n == 1:
10        return True
11    return is_even(n - 1)
12 try:
13     print('is_even(5)=', is_even(5))
14     print('is_odd(5)=', is_odd(5))
15 except RuntimeError:
16     print('StackOverflow')
17 EOF
18 $ python3 foo.py
```

Fraction of LLMs with correct answer: 12 / 16 = 0.75

Problem 4. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def binary_search(xs, y):
4     def go(lo, hi):
5         if lo > hi:
6             return False
7         mid = (lo + hi) // 2
8         if xs[mid] > y:
9             return go(lo, mid)
10        if xs[mid] < y:
11            return go(mid, hi)
12        return True
13    return go(0, len(xs) - 1)
14 try:
15     xs = [1, 3, 5, 7, 9]
16     print('binary_search(xs, 9)=', binary_search(xs, 9))
17 except RuntimeError:
18     print('StackOverflow')
19 EOF
20 $ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

LLM Model Performance

